THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The 50th odd number is: 1

(a) 100 **(b)** 99 **(c)** 101 **(d)** 97

2. The next number in the cube number sequence $1, 8, 27, 64, 125, \ldots$ is: 1

(a) 240 **(b)** 216 **(c)** 226 **(d)** 200

3. An angle of 45° is: 1

(a) obtuse **(b)** straight **(c)** right **(d)** acute

4. The number of degrees in half a turn is: 1

(a) 270 **(b)** 90 **(c)** 360 **(d)** 180

5. The estimate of 61 × 19 by rounding to tens is about: 1

(a) 1200 **(b)** 1000 **(c)** 600 **(d)** 1400

6. One step of the Kaprekar routine on 6321 gives: 1

(a) 5085 **(b)** 5094 **(c)** 5175 **(d)** 4995

7. The title of a graph should tell: 1

(a) only the date **(b)** what the graph is about
(c) the colours used **(d)** who drew it

8. A bar graph has bars of heights 12, 18, 15. The total is: 1

(a) 42 **(b)** 45 **(c)** 47 **(d)** 50

9. The LCM is the: 1

(a) least common multiple **(b)** least common factor
(c) largest common multiple **(d)** last common multiple

10. Which pair is a twin prime pair? 1

(a) 19 and 23 **(b)** 23 and 27 **(c)** 13 and 17 **(d)** 17 and 19

11. Perimeter of a square of side 9 m is: 1

(a) 18 m **(b)** 27 m **(c)** 81 m **(d)** 36 m

12. The area of a parallelogram-shaped figure can be found by converting it to a: 1

(a) line **(b)** rectangle **(c)** circle **(d)** cube

13. Which is greater? 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{4}$

14. Which is greater: $\frac{2}{3}$ or $\frac{3}{4}$? 1

(a) $\frac{2}{3}$ **(b)** $\frac{3}{4}$
(c) both are equal **(d)** cannot be compared

15. The diagonals of a rectangle: 1

(a) are perpendicular always **(b)** bisect each other
(c) never meet **(d)** are parallel

16. An arc is: 1

(a) a straight line **(b)** a triangle
(c) a corner **(d)** a part of a circle

17. The number of lines of symmetry of a regular pentagon is: 1

(a) 5 **(b)** 1 **(c)** 10 **(d)** 0

18. A kite has: 1

(a) 1 line of symmetry **(b)** no line
(c) 4 lines **(d)** 2 lines

19. The value of $-11 + (6)$ is: 1

(a) 5 **(b)** −7 **(c)** −3 **(d)** −5

20. $6 - 10$ equals: 1

(a) 16 **(b)** −4 **(c)** 4 **(d)** −16

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Write the next two terms: 1, 2, 3, 5, 8, … 2

22. What is the Collatz conjecture? 2

23. What are the axes of a bar graph? 2

24. What are co-prime numbers? Give an example. 2

25. What is the unit of area? Give an example. 2

26. Convert $3\frac{2}{5}$ to an improper fraction. 2

27. How is a square different from a rectangle? 2

28. Find $(-4) - (-9)$. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Write the Virahānka numbers up to 55. Which term is the sum of 13 and 21? 3

30. Find the angle between the hands at 2:00 and at 10:00. 3

31. The data 4, 7, 4, 9, 7, 4, 2 is given. Find the frequency of each value and the total. 3

32. Find the perimeter of a rectangle whose length is twice its breadth, with breadth 7 cm. 3

33. Explain with a drawing how the points equidistant from A and B are found. 3

34. Which of the words MOM, BOB, WOW, TOT read the same in a mirror placed vertically beside them? Give reasons. 3

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35. A rope of 48 m and another of 36 m are cut into equal pieces of the longest possible length. Find the length. 3

OR

A number is divisible by both 4 and 6. Is it divisible by 24? Explain with an example. 3

36. A bank account has ₹250. A cheque of ₹400 is issued. What is the balance? 3

OR

Find the integer which when added to −9 gives 4. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Why are tracks of a railway drawn parallel? 5
(ii) Give two examples of intersecting lines in surroundings.

OR

(i) Why do we use degrees for angles? 5
(ii) Why is one full turn 360°?
(iii) Find the angle turned in 1/6 of a turn.

38. (i) What is a 4-digit palindrome pattern? 5
(ii) Show that every 4-digit palindrome is divisible by 11.
(iii) Test with 3443.

OR

(i) What is a number line? 5
(ii) Mark 0, 250, 500, 750 and 1000 on a line.
(iii) Which of these is nearest to 640?

39. **Case study:** A tailor has 5 m of cloth. A shirt uses $1\frac{3}{4}$ m. 5
(i) How much for two shirts?
(ii) How much cloth is left?
(iii) Can he make a third shirt?
(iv) How much more is needed or left?

OR

(i) Compare $\frac{7}{3}$ and $2\frac{1}{2}$. 5
(ii) Add them.
(iii) Write the sum as a mixed fraction.

40. (i) Explain why a circle has infinite lines of symmetry. 5
(ii) Find its order of rotation.
(iii) How does it differ from polygons?

OR

(i) How are symmetry and reflection related? 5
(ii) Give two examples of rotational symmetry in daily life.

END OF QUESTION PAPER Total: 80 Marks

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